ALIFE 2026 · ORAL PRESENTATION

Conchordal

Emergent Harmony via Direct Cognitive Coupling in a Psychoacoustic Landscape

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MOTIVATION

Sound as an ALife substrate

Spatial	Chemical	Symbolic	Acoustic
cellular automata, morphogenesis	autocatalysis, artificial chemistry	genetic programming, rule systems	underexplored as a primary medium

THE SCIENTIFIC QUESTION

Can a landscape derived from **psychoacoustic observables** function as a non-traditional ALife substrate — supporting **self-organization, selection, heredity, and synchronization**?

Sound has rich physical structure and is deeply entangled with biological perception — yet it remains underexplored as a computational medium for life-like dynamics.

BEFORE THE LANDSCAPE

Why must it be this sound?

Pythagoras	Rameau	Schoenberg	Grisey · Murail	Takemitsu
6th c. BC	1722	1908	1970s	1990s
Harmony as whole-number ratios.	Harmony from the overtone series.	All twelve pitches made equal.	Composing from the measured spectrum.	"A sea of tonality" — the pull of sound itself.

Since Pythagoras, harmony has been sought in number. ***None of it answered the question.***

CORE PRINCIPLE

Direct Cognitive Coupling (DCC)

DESIGN PRINCIPLE

For perceptually grounded generative systems.

(i) Perceptual axes

Axes computed from perceptual observables: cochlear roughness, periodicity-based harmonicity.

(ii) Direct read-out

Agents read that landscape directly. No symbolic rules, no scales, no trained parameters.

WHAT THE PRINCIPLE CLAIMS

Ecological–Aesthetic Duality

One landscape governs survival and turnover. The same field is the proxy for musical coherence.

A claim about model structure. Human judgement not tested here.

WHAT THE ASSAYS TEST

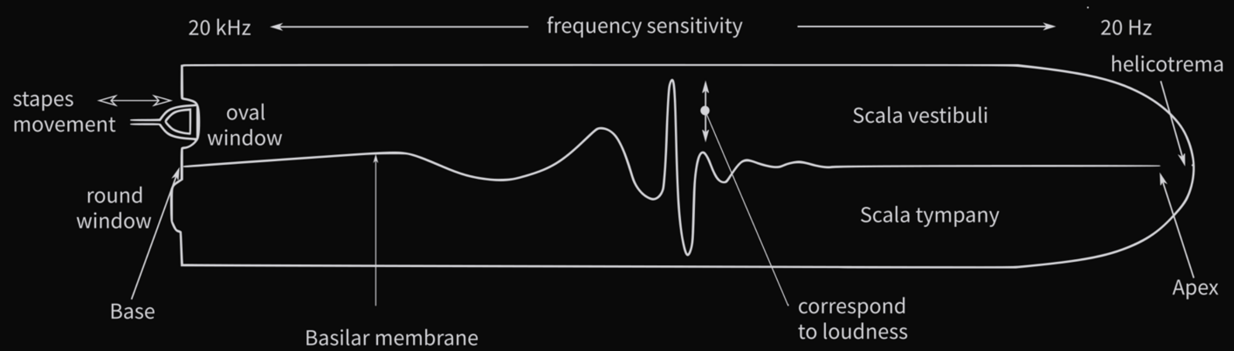
The scientific question

Can a landscape derived from **psychoacoustic observables** function as a non-traditional ALife substrate — supporting **self-organization, selection, heredity, and synchronization**?

The same question, now under test in four assays.

Place on the basilar membrane is frequency

The organ both landscape axes are read from.



Tonotopy

The membrane is stiff and narrow at the stapes end, wide and loose at the apex, so each place responds best to one frequency.

Why partials interact

Two partials close in frequency excite overlapping places. Inside one critical band the ear cannot separate them, and the residue is heard as roughness.

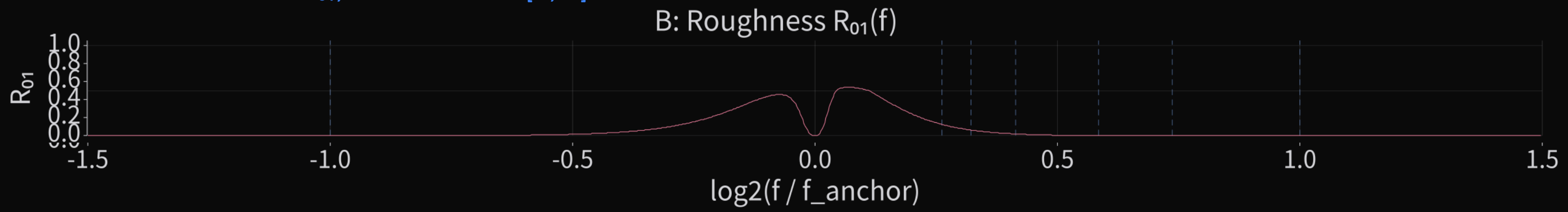
Why this matters here

The landscape's axes are not musical conventions. They are consequences of where sound lands on this membrane.

Roughness — beating inside the critical band

Sensory dissonance, computed from cochlear filtering.

THE LANDSCAPE AXIS · R_{01}, normalised to [0, 1]



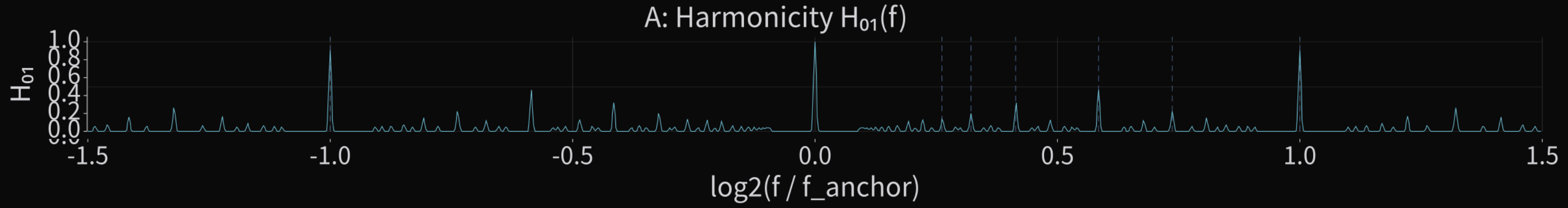
WHERE IT COMES FROM · early peripheral processing, rooted in cochlear mechanics

Troughs line up with simple ratios: there the interval is wide enough that non-coincident harmonics fall outside each other's critical bands. *Plomp & Levelt, 1965 · Sethares, 1993*

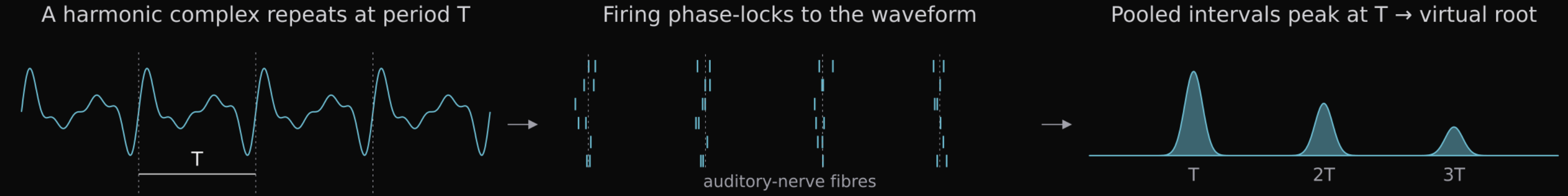
Harmonicity — evidence for a virtual root

Periodicity, computed against a harmonic template.

THE LANDSCAPE AXIS · H_{01}, two-pass sibling projection, $N = 16$, $\rho = 0.4$



WHERE IT COMES FROM · localized in the auditory brainstem and primary auditory cortex

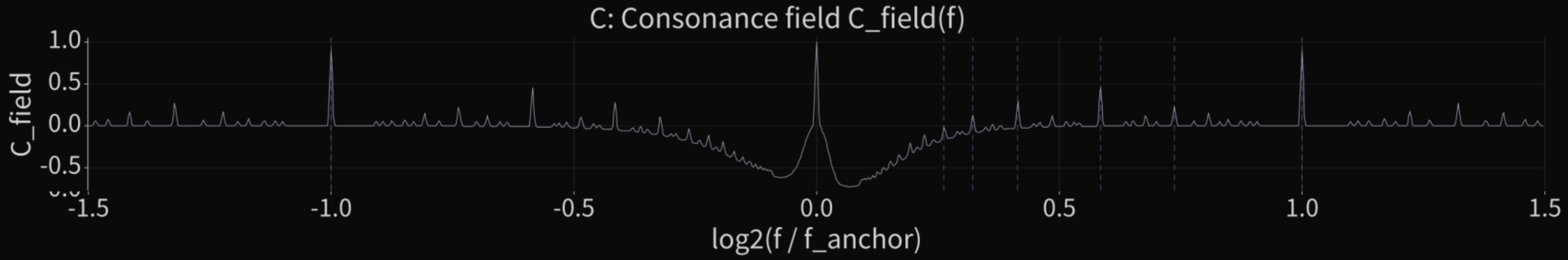


Peaks line up with simple ratios: coincident harmonics accumulate constructively, and the number of coincidences within order N grows as max(p, q) shrinks. *Terhardt, 1979* · *Parncutt, 1988*

Consonance — a bilinear core over the two axes

Derived like the axes it combines — in NK landscapes ruggedness is a tuning knob; here nobody set it.

THE LANDSCAPE AXIS · $C\_field = H_{01} - 1.35\ R_{01} + H_{01} \cdot R_{01}$, attractors near 2:1, 3:2, 4:3, 5:4



WHERE IT SITS · not one place — and that is why a formula has to be chosen

R **Cochlea** basilar membrane → **H** **Brainstem** periodicity extraction → **H** **Auditory cortex** virtual pitch → **C** **Higher-order** shaped by enculturation

$$C = a \cdot H_{01} + b \cdot R_{01} + c \cdot H_{01} \cdot R_{01} + d$$

the most general bilinear form for two inputs on [0, 1]; $b < 0$ penalises R

We built this landscape into an instrument

Conchordal, running scenario autumn_cycle

Conchordal is a software instrument, running in real time.

Its sounds are not placed notes. They are agents that live, move and die in this landscape.

The bottom panels are the quantities of the last three slides — the consonance landscape, roughness and harmonicity — computed live from the sound.

The assays that follow are its minimal core, cut out for measurement.

Three minimal mechanisms

Deliberately minimal, so the landscape's contribution is attributable.

1 Pitch adaptation

Local proposals, scored leave-one-out.

$$S = C_{field} - \lambda_C \cdot \Sigma K_{crowd} - \lambda_M |\Delta x|$$

Metropolis acceptance under a decaying temperature. The crowding kernel is the analytic complement of the roughness kernel.

2 Metabolic policy

Consonance acts as an ecological resource.

$$E \leftarrow E - c_b \Delta t + r_E \cdot \max(0, C) \cdot \Delta t$$

Burn at a fixed rate, recharge with consonance, die at zero.

3 Temporal oscillation

One phase oscillator per agent.

$$\dot{\varphi}_i = \omega_i + K \sin(\theta^*_i - \varphi_i)$$

Coupled to a shared field, not to each other. The interaction is bidirectional.

Each is a simplification of a full instrument subsystem. What is under test is the landscape, not the search strategy.

EXPERIMENTS

Four assays, four mechanism subsets

✓ = on, ✗ = off, abl. = ablated (compared with both states). All assays use a fixed 220 Hz reference.

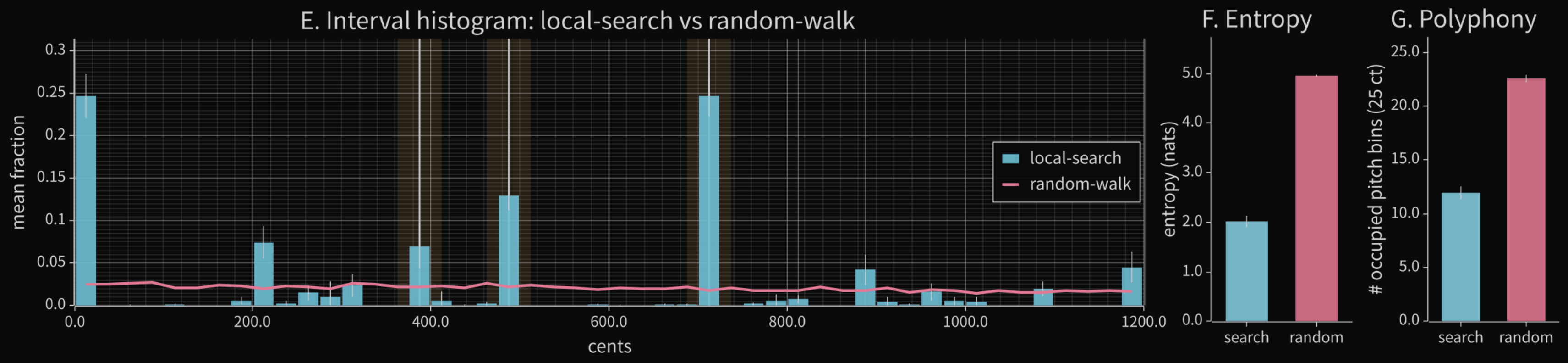
Condition	Search	Selection	Heredity	Entrainment
Agents N	24	32	16	32
Seeds	20	20	20	20
Hill climb	abl.	✗	✗	✗
Crowding	✓	✗	✓	✗
Metabolism	✗	abl.	abl.	✗
Entrainment	✗	✗	✗	✓
Heredity	✗	✗	abl.	✗
Duration	8 sweeps	200 deaths	12k steps	40 s

Each assay isolates a different mechanism subset.

EXPERIMENT 1 · CONSONANCE SEARCH

Hill-climbing yields structured polyphony

24 adaptive voices + a fixed 220 Hz drone on a 3 ct grid; hill-climbing vs. a matched random local walk.



INTERVAL ENTROPY (NATS)

2.01 vs **4.96**

Uniform ceiling 5.48.

UNIQUE PITCH BINS

11.95 vs **22.60**

24 agents settle into ≈ 12 distinct pitch classes.

JI PROXIMITY SCORE

0.366 vs **0.115**

Weighted proximity to simple ratios. Confirms the entropy result.

Each pair reads local-search first, then the matched random-walk ablation.

AUDIO DEMO 1

Self-organized polyphony

Same schedule, same step window, same pitch bounds — only the acceptance rule differs.

♪ Listen

local-search

Hill-climbing on the exact LOO consonance objective under the crowding penalty.

Click to play · 2 seeds × 8 s

random-walk ablation

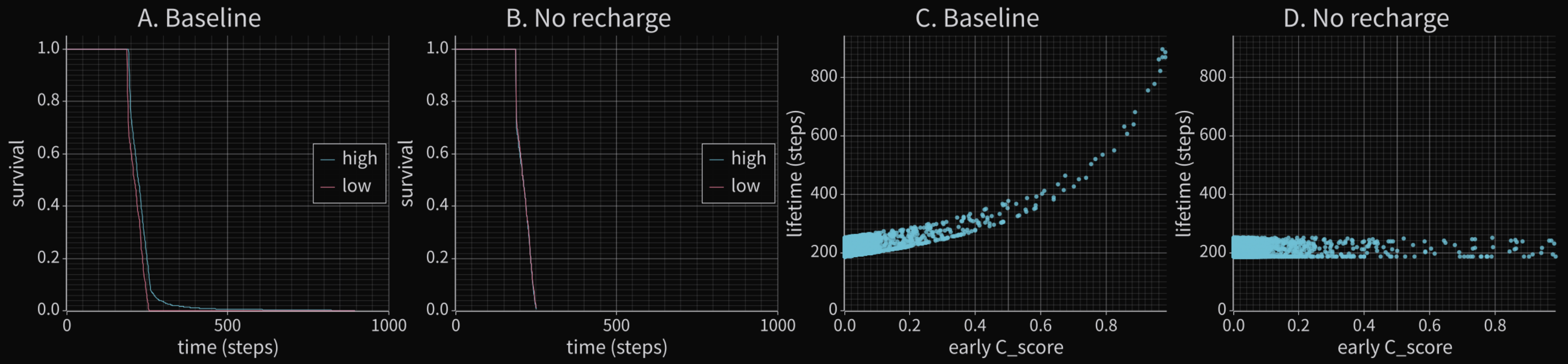
Matched random local walk replaces hill-climbing. Everything else identical.

Click to play · 2 seeds × 8 s

Two seeds × 8 s each, rendered directly from the simulation state. Audio is part of the paper's supplementary material.

Consonance becomes an ecological resource

32 agents within ±1 octave of the anchor; dead agents respawn at random until 200 deaths accumulate.



RECHARGE ON · SEED-LEVEL

$r = 0.718$

Early consonance predicts lifetime ($n = 20$, $p < 0.001$).

RECHARGE OFF · ABLATION

$r = 0.000$

Not different from zero. Pitch adaptation is off, so this is not positional coincidence.

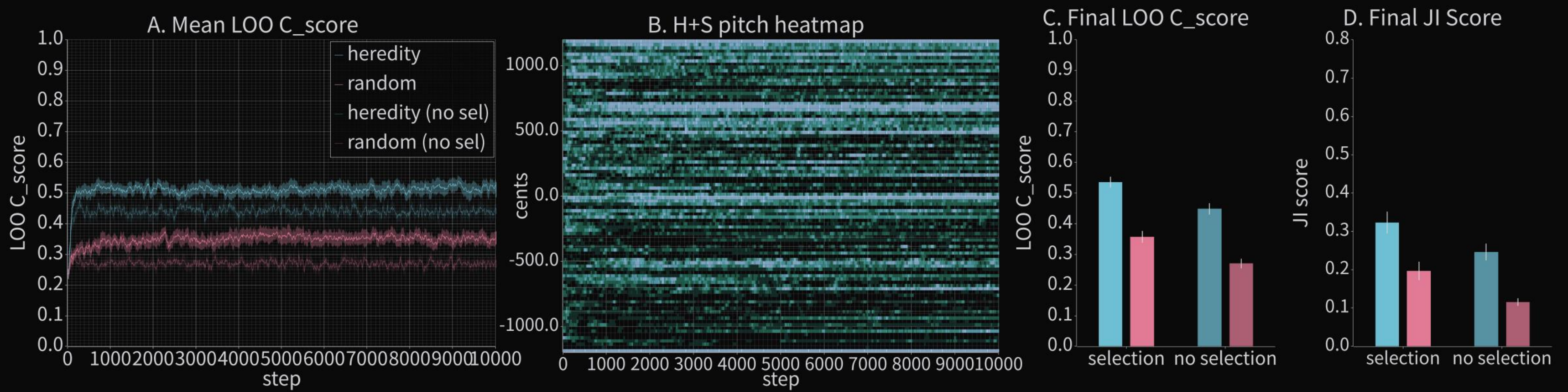
CONDITION CONTRAST

$t(23.2) = 18.7$

The differential requires consonance-dependent recharge.

Lineage bias accumulates across generations

2 × 2 factorial: respawn (heredity vs. matched log-random) × metabolic selection, adult pitch locked.



FINAL LOO C_SCORE

0.536 vs **0.358**

heredity+selection vs. matched random+selection. Heredity alone still wins (0.449 vs. 0.271).

JI PROXIMITY / ENTROPY

0.323 / 3.28

vs. 0.197 / 4.22 for matched random. Occupied bins contract 14.35 → 12.30: concentration, not scatter.

SEEDS CROSSING JI ≥ 0.50

15 / 20

No matched-random run crosses it. What works: family heredity + local pitch re-optimisation + selection.

AUDIO DEMO 2

Hereditary adaptation you can hear

Adult pitch is locked in both arms — the only difference is how newborns are placed.

♪ Listen

heredity + selection

Parent-biased family choice, local pitch re-optimisation, metabolic selection on.

Click to play · 12 s

matched random + selection

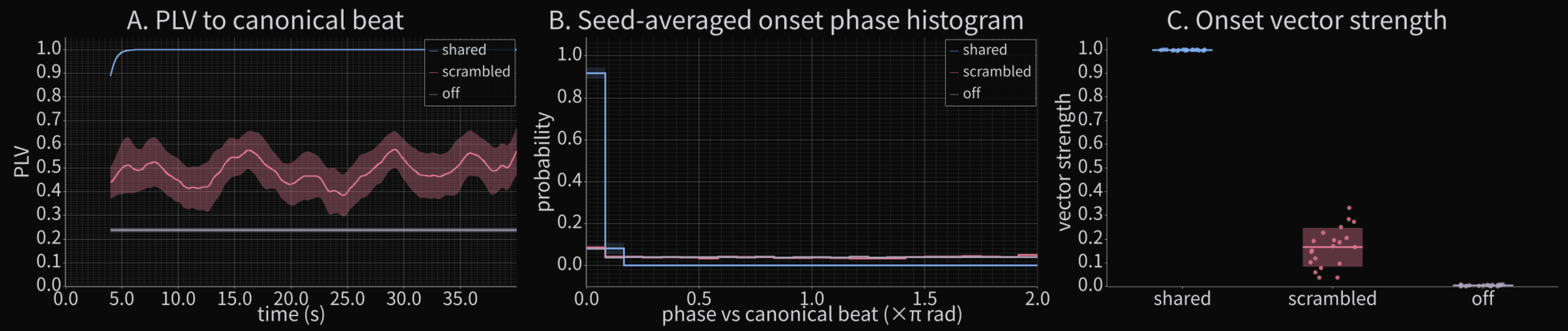
Log-random candidate sampling under the same scene filter; selection still on.

Click to play · 12 s

Only heredity+selection produces clearly musical multi-band polyphony; the controls remain diffuse. Informal listening, reported as such.

A shared field organizes timing

All agents reduced to phase oscillators, driven by an external 2 Hz scaffold.



SHARED DRIVE

0.998

All agents couple to one continuous 2 Hz phase reference.

SCRAMBLED DRIVE

0.166

Same cycle length and coupling, but phase re-randomized each cycle.

DRIVE OFF

0.005

Coupling disabled; intrinsic frequencies retained.

Expected under coupled-oscillator theory. Temporal-axis validation, not an emergence claim.

ROBUSTNESS

The landscape, or any scalar reward?

A random permutation of consonance bins, held fixed for the run.

Same distribution of scores. Only the spatial arrangement is destroyed.

Metric	Intact landscape		Shuffled	Effect (all $p < 0.001$)
Interval entropy (nats)	2.013	→	4.416	*structure destroyed*
Unique pitch bins	11.9	→	17.0	*crowding over-disperses*
Nearest-neighbour spacing (ct)	0.576	→	0.928	*no collapse, but no order*

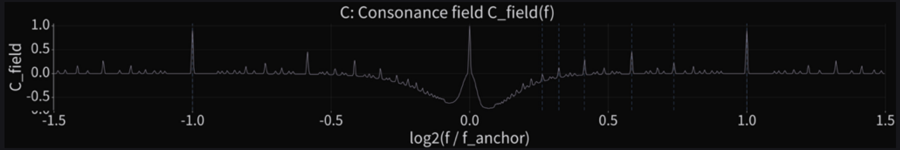
It is not the presence of a fitness signal that matters — a coherent psychoacoustic gradient is required.

Caveat: the hereditary assay has not been through this battery.

The ecological–aesthetic duality

In conventional generative music systems, ecological fitness and aesthetic evaluation are separate functions.

One psychoacoustic landscape



Ecological terrain

Survival, movement, turnover and synchronization are all landscape position.

Aesthetic proxy

The same field is the system's measure of musical coherence.

Roughness and crowding keep the system off unison — easy, and trivial.

What this does not yet show

The claims the present work does not reach.

No behavioural validation

The duality predicts that ecological success tracks perceptual coherence. No listener study has been run.

A Western-leaning profile

Weighting varies across listeners and cultures. These coefficients encode one profile.

McDermott et al., 2016 — Tsimane' listeners rated consonant and dissonant chords alike.

gamelan · maqam · shruti · gagaku shō · inharmonic timbre

A fixed reference

All assays fix a 220 Hz reference — a compositional choice, not a system constraint. Remove it and the emergent tuning itself can drift.

Not open-ended

The landscape moves with the agents, but the generating rules never change. The space of what can appear is closed. Evolvable timbres and coupling topologies are the routes onward.

CONCLUSION

A viable substrate for artificial life

Self-organisation	Selection	Heredity	Entrainment

A continuous landscape derived from harmonicity and roughness sustains the core ingredients of an artificial ecology **without symbolic encoding**.

Taken together: DCC-derived consonance is a viable ALife substrate, complementing the spatial and chemical media the field has traditionally studied.

DCC is not inherently auditory

What the principle asks for is a quantitative perceptual model, not a particular sense.

The same two requirements: **(i) Perceptual axes** · **(ii) Direct read-out**

Visual

Spatial frequency and colour-opponent structure are already modelled quantitatively. A terrain built from those would stand in for art-historical rules.

Tactile

Vibrotactile perception has its own roughness. The closest formal analogue to the axis here.

Kinesthetic

Effort, balance and timing admit quantitative description. A landscape of motor-perceptual cost.

Any creative medium whose perceptual basis admits quantitative modelling could host an analogous ecology.

TRY IT

Conchordal is a playable instrument

The assays here use minimal dynamics. Each is a simplification of a subsystem in the full instrument.

For composers, it offers a generative substrate in which ecological dynamics produce musically coherent material — without symbolic specification.

Acknowledgment

The author thanks Gilberto Bernardes for helpful correspondence regarding the naming of Conchordal in relation to his Conchord system.

BETA TESTERS WELCOME

Composers · sound artists · ALife researchers

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